Interior Closure Termination and the Structural Inadmissibility of Gravitational Singularity in Darke Theory

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2026

Abstract

This paper closes the remaining gravitational leg of the Darke Theory anti-pathology programme. Earlier work already established two independent realised instances of the same structural law: nontrivial finite-energy stationary matter cannot collapse to a point or disperse indefinitely, and a pre-threshold origin regime cannot be realised as a singular geometric origin. The present paper extends that law into gravitational extreme regimes.

The central claim is exact. If geometry is emergent rather than primitive, if geometric validity is threshold-conditioned, and if persistent realised regimes require finite support, then a bounded gravitational domain cannot admit continued geometric evolution into a divergent interior singularity. The reason is structural. Divergent interior continuation requires unbounded geometric burden while the support available to the bounded regime remains finite and, under intensifying closure, strictly decreases. The singular interior is therefore not the stable realised form of the regime.

The horizon is accordingly reinterpreted as a closure-saturation boundary: the minimal surface at which the burden required for continued recoverable geometric description equals the admissible support of the regime. Interior to that boundary, closure becomes dominant, admissible future geometric continuation collapses, and ordinary exterior-style geometric recoverability ceases to remain structurally complete. The singularity is therefore replaced not by a decorative smoothing, but by interior closure termination.

This construction unifies three previously separated Darke-theoretic results: the coherence-burden anti-pathology law, the finite-support relativistic closure bridge, and the terminal-closure end-state model. The same structural principle governs matter, cosmological origin, black-hole confinement, and cosmic finality: reality does not sustain regimes whose required persistence burden exceeds admissible support.

1. Introduction

General relativity is one of the great achievements of theoretical physics, but its strongest regimes carry a known wound. Continued classical extrapolation of gravitational collapse drives curvature and density toward singularity. The standard response has usually taken one of two forms. Either the singularity is accepted as the formal endpoint of classical geometry, or it is deferred to a future quantum-gravity completion.

Darke Theory allows a harder answer.

Across the Darke Theory corpus, pathological limits are not rejected because they are unpleasant, nor because they are dynamically inconvenient, but because they demand support that the governing regime cannot carry. This law has already been realised in two independent domains. In the matter-sector of the Darke Theory Field Equation, nontrivial finite-energy stationary matter cannot collapse to a point and cannot disperse indefinitely; finite preferred scale is forced. In the cosmological origin setting, a pre-threshold regime cannot be realised as a singular geometric origin because singular geometric description presupposes geometry where geometry is not yet admissible.

The remaining gravitational question is therefore immediate. If geometry is emergent rather than primitive, and if support is finite rather than infinite by fiat, can a bounded gravitational domain continue as geometry all the way to a singular interior?

The present paper answers no.

The proof does not rely on modifying commutators, adding ad hoc cutoffs, or importing external singularity cures. It proceeds directly from the internal architecture already established elsewhere in the corpus:

1. geometry is activated only above closure threshold;
2. persistence requires finite support;
3. admissibility filters regimes by support versus burden;
4. increasing closure compresses admissible future reorganisation;
5. black holes are confinement regimes whose horizon already marks support saturation.

Once these are combined, continued singular interior evolution is no longer a deep mystery. It is structurally inadmissible.

The horizon is therefore not merely a kinematic surface. It is the admissibility boundary at which the burden required for continued recoverable geometric description saturates the support budget of the regime. Interior to that boundary, closure becomes dominant, admissible future geometric continuation collapses, and singular extrapolation ceases to describe the realised regime.

This paper has one purpose: to state that theorem cleanly.

2. Foundational Setting

Only the minimum load-bearing Darke-theoretic commitments are required here.

Definition 1 (Geometric admissibility)**.** *Geometry is admissible if and only if the coherence functional exceeds critical threshold:*

$$\exists G \iff \Phi[C] > \Phi_{\text{crit}}. \tag{1}$$

Definition 2 (Pre-threshold regime)**.** *A regime is pre-threshold if*

$$\Phi[C] \leq \Phi_{\text{crit}}, \tag{2}$$

in which case geometric description is not admissible.

Definition 3 (Structural admissibility)**.** *A regime is structurally admissible if it can persist within the closure, persistence, and refinement constraints of the governing system without requiring unsupported burden.*

Definition 4 (Pathological limit)**.** *A pathological limit is a limiting attempted realisation of a regime whose persistence either destroys the support required by that regime or presupposes support not available within that regime.*

Principle 1 (Coherence-Burden Principle)**.** *If the coherence burden required for persistence exceeds admissible support, the regime is not structurally admissible as a stable realised form.*

Remark 1. *This principle is already realised in the anti-pathology theorem for matter and cosmological origin. The present paper extends that same law into the gravitational interior regime.*

3. Finite Support and the Geometric Regime

The relativistic closure–curvature bridge already fixed the governing posture of the geometric sector.

Axiom 1 (Threshold activation)**.** *Geometry is activated only in the post-threshold regime.*

Axiom 2 (Finite support)**.** *Persistence of a regime requires finite support and is not granted unbounded carrying capacity by default.*

Remark 2. *Finite support is not an independent postulate introduced for this result. It follows from the closure-based ontology in which persistence is realised only through bounded structural support. This constraint is already enforced in the relativistic closure–curvature bridge and in the coherence-burden framework.*

Axiom 3 (Admissibility filter)**.** *Only configurations with nonnegative total admissibility can qualify as persistent regimes.*

Remark 3. *These are not optional additions for the present paper. They are already built into the bridge construction. The present theorem uses them exactly as stated.*

Definition 5 (Bounded geometric domain)**.** *A bounded geometric domain is a post-threshold region*

$$\Omega \subset M$$

with finite spacetime support over the persistence horizon relevant to the regime.

Definition 6 (Geometric support)**.** *For a bounded geometric domain Ω, geometric support is the total admissible carrying capacity available to sustain recoverable geometric description across the required persistence horizon.*

Lemma 1 (Bounded domains do not possess unbounded support)**.** *Let Ω be a bounded geometric domain in the post-threshold regime. Then the support available to Ω is finite.*

Proof. Finite support is an axiom of the bridge construction. A bounded domain is not granted unbounded carrying capacity merely because geometric fields can be formally written on it. Therefore the support budget associated with Ω is finite. □

4. Closure Accumulation and Admissible Future Compression

The end-state paper established the global form of a deeper structural law: increasing closure compresses admissible future reorganisation until it vanishes under terminal closure. The same logic can be stated locally.

Definition 7 (Local closure density)**.** *Let $C(x)$ denote the effective closure density carried by a bounded domain at location x.*

Definition 8 (Local reorganisational capacity)**.** *Let R_Ω denote the effective reorganisational capacity of a bounded domain Ω, understood as the structurally meaningful admissible future continuations available to that domain.*

Principle 2 (Closure accumulation compresses admissible futures)**.** *If closure density increases without a compensating mechanism that opens new lawful structural freedoms, then the admissible future state space of the regime contracts in its meaningful extent.*

Lemma 2 (Local closure compression)**.** *Let Ω be a bounded post-threshold geometric domain. If closure density within Ω increases monotonically under sustained confinement, then R_Ω decreases.*

Proof. This is the local bounded-domain form of the closure-compression law already proved at the cosmic end-state level. Closure is restrictive as well as productive. When closure accumulates inside a bounded domain, admissible future reorganisation is compressed rather than enlarged. Therefore R_Ω decreases as closure density intensifies, unless offset by a compensating lawful freedom-opening mechanism. No such mechanism is assumed here. □

Corollary 1 (Interior terminalisation tendency)**.** *In a bounded domain undergoing sustained confinement-driven closure accumulation, admissible future geometric continuation tends toward exhaustion.*

5. Black Holes as Terminal Confinement Regimes

The standalone black-hole paper already redefined the black hole as a confinement regime rather than primarily a singularity object.

Definition 9 (Burden density)**.** *Let*

$$b(x,t) := \alpha\hat{\rho}(x,t) + \beta\hat{K}(x,t) + \gamma\hat{\Theta}(x,t), \tag{3}$$

where $\hat{\rho}$, $\hat{K}$, and $\hat{\Theta}$ are normalised density, curvature-burden, and compressive-collapse contributions, and $\alpha, \beta, \gamma > 0$.

Definition 10 (Support density)**.** *Let*

$$s(x,t) := \Phi_{\mathrm{loc}}(x,t) = \frac{1}{1+b(x,t)}. \tag{4}$$

Definition 11 (Integrated burden and support)**.** *For a spacetime region R over persistence horizon τ, define*

$$B(R;\tau) := \int_0^\tau \int_{R(t)} b(x,t)\, dV_t\, dt, \tag{5}$$

and

$$S(R;\tau) := \int_0^\tau \int_{R(t)} s(x,t)\, dV_t\, dt. \tag{6}$$

Definition 12 (Confinement functional)**.** *The confinement functional is*

$$\Lambda(R;\tau) := \frac{B(R;\tau)}{S(R;\tau)}, \qquad S(R;\tau) > 0. \tag{7}$$

Definition 13 (Structural horizon)**.** *The structural horizon is the minimal radius r_h for which the confinement ratio saturates the admissibility boundary:*

$$\Lambda(r_h) = 1, \qquad \Lambda(r) > 1 \text{ for } 0 \le r < r_h. \tag{8}$$

Remark 4. *This definition already encodes the key reversal. The horizon is not merely where escape fails. It is where the burden required for continued recoverable exterior geometric description reaches the support limit of the regime.*

6. Singular Interior Continuation as a Support Problem

We now isolate the exact object to be excluded.

Definition 14 (Singular interior continuation)**.** *A singular interior continuation is a purported continuation of a bounded post-horizon gravitational regime in which:*

1. *geometric description remains admissible throughout the interior;*
2. *curvature burden diverges along the continuation;*
3. *the continuation is treated as the stable realised form of the interior regime.*

Lemma 3 (Divergent interior continuation requires unbounded burden)**.** *Let Ω be a bounded post-horizon interior domain. If curvature burden diverges under a continued geometric interior description, then the total burden required for that continuation is unbounded.*

Proof. By definition, burden density includes a strictly positive curvature-burden term. If curvature burden diverges while geometric continuation is maintained, then the integrated burden cannot remain finite under the purported continuation. Therefore the total burden required for that continuation is unbounded. □

Lemma 4 (Support does not increase under closure-dominant confinement)**.** *Let Ω be a bounded post-horizon interior domain undergoing closure-dominant confinement. Then the support available to sustain recoverable geometric continuation is a strictly decreasing function of burden density once confinement is active.*

Proof. From the constitutive support law

$$s(x,t) = \frac{1}{1 + b(x,t)},$$

support density is a strictly decreasing function of burden density. Under closure-dominant confinement, density, curvature load, and compressive demand intensify rather than relax. Therefore support density does not increase and, once burden rises, strictly decreases. Since the domain is bounded, the total support budget remains finite and contracts under sustained burden growth. □

Corollary 2 (Interior support-collapse tendency)**.** *Inside the structural horizon, increasing confinement drives the regime toward decreasing support and increasing burden simultaneously.*

7. Interior Closure Termination Theorem

We can now state the main result cleanly.

Theorem 1 (Interior Closure Termination Theorem)**.** *Let Ω be a bounded post-threshold gravitational domain lying interior to the structural horizon of a collapse regime. Assume:*

1. *geometry is threshold-conditioned rather than primitive;*
2. *persistence requires finite support;*
3. *closure accumulation compresses admissible future continuation;*
4. *support density decreases under increasing burden density;*
5. *singular interior continuation requires divergent geometric burden.*

Then no singular interior continuation can be structurally admissible as the stable realised form of the regime.

Proof. Let Ω be bounded and post-horizon. By finite-support discipline, the total support available to Ω is finite. By closure-dominant confinement, burden density increases while support density decreases. Therefore the domain evolves into a regime in which the support budget contracts rather than expands.

Suppose, for contradiction, that a singular interior continuation is structurally admissible as the stable realised form of Ω. By definition of singular continuation, geometric description remains valid while curvature burden diverges. By the previous lemma, the total burden required for that continuation is therefore unbounded.

But the support available to the bounded regime is finite and decreasing. Hence the burden required for the purported continuation exceeds admissible support. By the coherence-burden principle, the continuation is not structurally admissible.

This contradicts the assumption that the singular interior continuation is the stable realised form of the regime.

Therefore no singular interior continuation can be structurally admissible as the stable realised form of the bounded post-horizon gravitational domain. □

Corollary 3 (Structural inadmissibility of gravitational singularity)**.** *A gravitational singularity is not the stable realised form of a bounded black-hole interior in Darke Theory.*

Corollary 4 (Interior termination before divergence)**.** *In a bounded black-hole interior, admissible geometric continuation terminates before any purported singular divergence can qualify as the realised regime.*

Remark 5. *The exclusion established here is structural, not dynamical. The result does not depend on the detailed field equations of general relativity or on any particular quantum correction scheme. It follows from admissibility alone.*

8. The Horizon as Closure-Saturation Boundary

The previous theorem changes the ontological role of the horizon.

Theorem 2 (Horizon Closure-Saturation Theorem)**.** *The structural horizon is the minimal surface at which the burden required for continued recoverable geometric description equals the admissible support available to the regime. Interior to that surface, ordinary exterior-style recoverable geometry is no longer structurally complete.*

Proof. This is immediate from the confinement criterion. The horizon is defined by

$$\Lambda(r_h) = 1.$$

At that surface, required burden equals available support. Interior to it, $\Lambda(r) > 1$, so burden exceeds support. By the coherence-burden principle, continued recoverable exterior-style geometric description cannot remain structurally admissible. Thus the horizon is the closure-saturation boundary of the recoverable geometric regime. □

Remark 6. *This does not mean the interior is unreal, empty, or null. It means the interior cannot be carried as a continued ordinary recoverable geometric regime of the same kind as the exterior. The singularity is therefore replaced by closure-dominant incompleteness of the classical descriptive basis.*

9. Relation to the Coherence Boundary and Evaporation

The strict coherence-boundary evaporation paper already established that geometry exists only for

$$\Phi(M) \geq 1,$$

and terminates at the Planck threshold rather than continuing into a divergent geometric endpoint.

The present theorem is the interior counterpart of that endpoint law.

Proposition 1 (Two-sided geometric termination)**.** *Darke Theory now supplies two structurally matched termination laws for black holes:*

1. *interior geometric continuation terminates before singular divergence because burden exceeds support in the bounded confinement regime;*
2. *exterior geometric thermodynamics terminates at the coherence boundary $M = M_P$, below which geometry is not admissible.*

Proof. The first statement is exactly the Interior Closure Termination Theorem proved above. The second is the strict coherence-boundary evaporation result. Together they show that neither the deep interior nor the evaporation endpoint admits unrestricted divergent geometric continuation. □

10. Local Terminal Closure and the Global End-State

The end-state paper established that a universe reaches terminal closure when admissible future reorganisation collapses to zero under globally dominant closure. The present theorem shows that black-hole interiors are the local bounded-domain version of the same law.

Theorem 3 (Local–Global Terminal Closure Correspondence)**.** *Terminal closure is the same structural law at two scales:*

1. *globally, it is the asymptotic exhaustion of admissible future reorganisation for the universe;*
2. *locally, it is the exhaustion of admissible future geometric continuation inside a bounded closure-dominant confinement domain.*

Proof. In the global case, closure accumulation compresses the admissible future of the universe until reorganisational capacity tends to zero. In the local black-hole case, closure accumulation inside a bounded confinement domain compresses admissible future geometric continuation until ordinary recoverable geometry can no longer be carried. The scale differs, but the structural logic is identical: increasing closure compresses admissible future until continuation fails. □

Remark 7. *This is the unification point. Origin, matter, black holes, and cosmic finality are no longer separate metaphysical stories. They are governed by the same admissibility law.*

11. Formation-to-Termination Continuity of Black Hole Admissibility

The preceding sections established three independent structural facts:

1. black hole formation occurs at confinement saturation, defined by

$$\Lambda(r_h) = 1; \tag{9}$$

2. singular interior continuation is structurally inadmissible, since burden diverges while support remains finite and contracts;
3. geometric description terminates at the coherence boundary

$$\Phi(M) = 1, \tag{10}$$

   below which geometry is not admissible.

Taken individually, these statements describe formation, interior behaviour, and evaporation endpoint. The present section shows that they are not independent phenomena, but scale-specific realisations of a single admissibility law.

11.1. Two Threshold Forms of One Law

The confinement functional

$$\Lambda = \frac{B}{S}$$

measures the ratio of burden to support within a bounded domain. Formation occurs when burden saturates support at the boundary:

$$\Lambda(r_h) = 1.$$

The coherence functional

$$\Phi$$

measures the degree of closure required for geometric admissibility. Geometry exists only when

$$\Phi > \Phi_{\text{crit}},$$

and terminates at

$$\Phi = \Phi_{\text{crit}}.$$

In the evaporation-sector normalisation used here, the critical geometric threshold is fixed by convention to

$$\Phi_{\text{crit}} = 1.$$

These functionals operate at different scales:

- Λ governs local confinement and persistence;
- Φ governs global geometric admissibility.

They are not independent. Both encode the same structural constraint: persistence requires that the burden imposed by a regime does not exceed the support available to carry it.

11.2. Lifecycle Structure of a Black Hole

A black hole therefore realises the coherence-burden law across a complete lifecycle.

Formation. Collapse drives increasing burden density. A black hole forms precisely when the boundary of the collapsing region reaches

$$\Lambda = 1,$$

so that the burden required for continued recoverable exterior geometry saturates available support.

Interior evolution. Within the horizon, confinement continues to increase burden density while support density decreases. The domain is bounded, so total support remains finite. Any continued geometric evolution that drives curvature toward divergence therefore requires unbounded burden within a finite-support system. By the coherence-burden principle, such continuation is inadmissible. Interior evolution therefore proceeds toward closure termination rather than singular divergence.

Evaporation and endpoint. As the system evolves, geometric support decreases globally. The geometric regime terminates when

$$\Phi = \Phi_{\text{crit}},$$

at which point geometry itself is no longer admissible. The endpoint is therefore not a divergent remnant, but coherence-boundary termination of geometric description.

11.3. Lifecycle Continuity Theorem

Theorem 4 (Black Hole Lifecycle Admissibility)**.** *Formation, interior evolution, and geometric termination of a black hole are governed by a single coherence-burden admissibility law. Specifically:*

1. *formation occurs at confinement saturation ($\Lambda = 1$);*
2. *interior singular continuation is inadmissible because it requires unbounded burden within finite support;*
3. *geometric description terminates at the coherence boundary ($\Phi = \Phi_{\text{crit}}$).*

These are not independent mechanisms but scale-specific realisations of the same support-versus-burden constraint.

Proof. Formation is defined by the condition that the burden required for continued exterior recoverable geometry equals the support available to the regime, expressed by $\Lambda(r_h) = 1$.

Interior evolution occurs within a bounded domain. By finite-support discipline, available support is finite. Under confinement, burden density increases while support density decreases. Singular continuation requires divergent burden. Singular continuation therefore requires divergent burden within a bounded finite-support domain. Hence the required burden exceeds the admissible support budget of the regime, violating admissibility.

Termination occurs when geometric admissibility fails, defined by $\Phi = \Phi_{\text{crit}}$. Below this threshold, geometry is not a valid descriptive regime.

In all three cases, the same condition holds: persistence fails when required burden exceeds available support. The formation threshold, interior termination, and geometric endpoint are therefore expressions of one structural law. □

11.4. Continuity with Global Terminal Closure

This lifecycle structure is the bounded-domain analogue of the global terminal closure law. At the cosmic scale, closure accumulation compresses admissible future reorganisation until it vanishes. In the black-hole case, the same process occurs locally: closure accumulation compresses admissible geometric continuation until it terminates.

Corollary 5 (Local–Global Admissibility Continuity)**.** *Black hole lifecycle behaviour and cosmic end-state evolution are governed by the same closure-driven admissibility constraint, differing only in scale and domain.*

11.5. Interpretive Consequence

A black hole is not correctly understood as a system that forms under one mechanism, evolves under another, and terminates under a third. It is a single regime whose entire lifecycle is governed by one law:

> Persistence is admissible only while required burden does not exceed available support.

Formation, interior behaviour, and evaporation are therefore not separate phenomena, but a continuous realisation of the same structural constraint.

12. Interpretive Consequences

The theorems above imply the following.

1. Black holes are not fundamentally singularity objects. They are terminal confinement regimes.
2. The horizon is not merely a geodesic escape surface. It is an admissibility boundary.
3. The interior is not best interpreted as a literal realised infinity. It is a closure-dominant phase in which classical descriptive completeness fails.
4. The evaporation endpoint is not divergent disappearance. It is coherence-boundary termination.
5. The same law that excludes pointlike matter and singular cosmological origin also excludes singular gravitational interior continuation.

None of these consequences depends on rhetorical inflation. They follow from the proved support-versus-burden structure.

13. Conclusion

This paper closes the remaining gravitational leg of the Darke Theory anti-pathology programme.

The earlier corpus had already established that pathological limits fail when the support required for their persistence cannot be carried by the governing regime. Matter provided one realised theorem. Cosmological origin provided another. The present paper extends the same law into gravitational interior regimes.

If geometry is threshold-conditioned rather than primitive, if persistence requires finite support, if closure accumulation compresses admissible future continuation, and if singular continuation requires divergent burden inside a bounded domain, then singular interior continuation cannot be the stable realised form of a black-hole interior.

The singularity is therefore not cured by decoration, smoothing language, or speculative patching. It is structurally inadmissible.

The correct replacement is interior closure termination. The horizon is the closure-saturation boundary at which recoverable geometry reaches its support limit. Interior to that boundary, closure becomes dominant, admissible future geometric continuation collapses, and classical descriptive completeness fails before any literal singular infinity can qualify as realised structure.

The result is stronger than an adjustment to black-hole interpretation. It completes a common law across four domains:

1. finite-scale matter,

2. non-singular cosmological origin,
3. non-singular gravitational interior,
4. terminal closure at cosmic finality.

The same principle governs them all. Reality does not sustain regimes whose required persistence burden exceeds admissible support.

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